

TIANYIN (TIM) QIU

Computational Chemist · Machine Learning & AI-Driven Drug Design (CADD / AIDD)

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SUMMARY OF QUALIFICATIONS

- **Drug Discovery & Design:** PhD-trained computational chemist (Yale) delivering structure-based and AI-driven drug design across small molecules, peptides/macrocycles, and bioconjugates — pairing physics-based methods (SBDD, FEP, MD, DFT) with machine learning to drive closed-loop, design–make–test–analyze (DMTA) optimization and medicinal-chemistry prioritization with SAR, ADME, and developability in mind.
- **AI / ML & Generative Modeling:** Build and embed modern AI directly into discovery — generative & diffusion models (RFdiffusion, ProteinMPNN), LoRA-fine-tuned LLM ensembles (Nature 2026), and agentic, LLM-orchestrated tool-use workflows — with active-learning, generate–score design loops that turn emerging methods into working pipelines fast.
- **Engineering & Delivery:** Ship production-grade, reproducible pipelines and internal tools rapidly (Python/PyTorch, Docker, Git) with AI coding assistants (Claude Code) on Slurm-managed, GPU-accelerated HPC.
- **Client-Facing & Communication:** Deliver clear technical rationale and method education to clients and medicinal chemists; strong communicator with high-impact publications (incl. Nature) and international conference presentations.
- **Leadership & Collaboration:** Lead Atombeat's agentic research function, project-manage a lead RDC program, and enabled Atombeat's first North America commercial client program; collaborate closely across computational and experimental teams and mentor undergraduate and graduate students.

TECHNICAL SKILLS

- **Agentic AI & LLM Workflows:** Multi-agent orchestration and agent-framework design (Claude Agent SDK, custom in-house orchestration), MCP (Model Context Protocol) servers and custom tool APIs for scientific software, agentic tool-use workflows driving OpenMM, Schrödinger, and other computational-chemistry tools, autonomous design–score–triage loops, LLM fine-tuning (LoRA / PEFT), model ensembling, retrieval-augmented literature & patent mining, prompt engineering, AI-assisted development (Claude Code).
- **Machine Learning & Generative Modeling:** Deep learning (PyTorch), generative & diffusion models (RFdiffusion, ProteinMPNN), protein language models (ESM), supervised & representation learning, molecular featurization / descriptors, multi-objective & QSAR / property prediction, active-learning closed-loop (generate–score) design cycles, model ensembling.
- **Drug Discovery & Design:** Computer-Aided & AI-Driven Drug Design (CADD / AIDD), Structure-Based Drug Design (SBDD), target assessment & druggability, hit discovery, lead optimization, closed-loop (design–make–test–analyze) optimization, high-throughput virtual screening (HTVS, billion-scale), de novo protein & binder design, co-folding & structure prediction (AlphaFold2/3, ColabFold, ESMFold), pharmacophore modeling, SAR / QSAR, ADME & developability, cheminformatics, peptide & macrocycle design, bioconjugates (e.g. RDC).
- **Modeling & Simulation:** Molecular Dynamics (MD), Enhanced Sampling, Free Energy Calculations (MM-PBSA, MM-GBSA, FEP, FEP+, RBFE), Docking, Quantum Chemistry / DFT, allosteric & community network analysis, protein–protein and protein–ligand interaction analysis.
- **Programming & HPC:** Python (PyTorch, Hugging Face, scikit-learn, Pandas, NumPy, Matplotlib, Jupyter), Linux/Bash, Fortran, CUDA; HPC: Slurm job scheduling, MPI / MPI4py parallelization, GPU acceleration, Docker containerization, Git.
- **Tools & Software:** Schrödinger Suite, Maestro, Glide, FEP+, AutoQSAR, AutoDock Vina, GNINA, GROMACS, AMBER, CHARMM, NAMD, OpenMM, MDAnalysis, MDTraj, AlphaFold, ColabFold, ESMFold, RFdiffusion, ProteinMPNN, PyRosetta / ROSETTA, RDKit, Prolif, VASP, Gaussian, PySCF, ASE, GFN-xTB, CREST, CCP4, cctbx; Claude Agent SDK, Claude Code, MCP servers.

PROFESSIONAL & RESEARCH EXPERIENCE

Senior AI & CADD Scientist, Agentic Research | Atombeat Inc., RiDYM0 Dept. | Remote

Oct 2025 – Present

Agentic AI Platform Development and Client-Facing Lead Optimization

- Lead and project-manage a client-facing lead-optimization program for a radioconjugate (RDC) targeting a transmembrane GPCR — driving FEP-based optimization of peptide potency and tumor uptake, coordinating cross-functional deliverables, and delivering computational data and design guidance on a regular cadence.

- Established and now lead Atombeat's agentic research function (since Jan 2026), building an in-house orchestration layer on the Claude Agent SDK that exposes internal computational-chemistry tools — structure preparation, docking, MD, FEP, and downstream analysis — as agent-callable services through MCP servers and custom tool APIs; the layer is now the default workflow substrate for discovery programs company-wide.
- Compressed protein–ligand structure preparation and FEP map setup from 3–5 days of hands-on work to a single unattended overnight run (<12 h), with agents resolving protonation and tautomer states, building and equilibrating systems, constructing perturbation maps, and submitting jobs for scientist review.
- Built autonomous design agents that close the generate–score–triage loop over peptide and small-molecule series, returning ranked design packages with structural rationale and interaction analysis rather than raw scores, and flagging edges and poses that need human adjudication.
- Led target assessment for the program's transmembrane GPCR and an early-stage oncology target — evaluating druggability, binding-site tractability, and design feasibility, and mining the literature and patent landscape with specialized LLM toolkits to surface prior art and reduce the client's patent-conflict risk.
- Designed and implemented an end-to-end, active-learning discovery platform for orally accessible cyclic peptides, integrating non-natural amino acids to improve permeability and stability; a closed-loop generate–score cycle optimizes predicted affinity across multiple objectives, achieving ~10% experimental hit rates in pilot campaigns.
- Developed a parallelized, RFDiffusion-centered de novo binder-design pipeline that produced over 100 structurally diverse candidates for an inflammatory bowel disease target, advancing multiple designs to experimental validation.
- Built and iteratively refined a high-fidelity protein–ligand model of the transmembrane GPCR through an agentic, LLM-orchestrated tool-use workflow — prompt-engineered to drive OpenMM and other computational-chemistry tools across multi-hour automated cycles — combined with residue-by-residue SAR interpretation.
- Reached an FEP mean absolute error of 0.85 kcal/mol against experiment across a ~50-compound linear-peptide series — high accuracy for FEP — yielding multiple validated candidates advanced to tumor-uptake follow-up studies.
- Carry the program's full computational workload single-handedly on this infrastructure, covering scope that conventionally staffs 3–5 CADD scientists.
- Accelerated platform delivery using AI coding assistants (Claude Code) to rapidly build and iterate internal discovery tools and pipelines.
- Produced the computational analyses, figures, and slide content presented by senior leadership at international conferences, including Drug Discovery Chemistry (DDC) and the ACS National Meeting.
- Turned the agent platform into part of the client deliverable: partners receive agent-generated design packages and structural analyses on a standing cadence, differentiating Atombeat's external R&D service model.
- Secured the company's first commercial contract by contributing technical strategy and scientific planning for a lead optimization collaboration with a publicly traded U.S. pharmaceutical partner, establishing Atombeat's external R&D service model.

Computational Modelling Intern | Boehringer Ingelheim | Ridgefield, CT

Jun 2024 – Aug 2024

Chemical Development — Machine Learning Driven High-Throughput Catalyst Design

- Accelerated rapid synthesis of drug compounds by 3–4 weeks via a fully automatic training pipeline testing over 10,000 machine learning models, leveraging 28 DFT-derived descriptors for accurate selectivity predictions.
- Automated the prediction of reaction outcomes across 3 metals and 2 ligand families with minimal manual intervention; achieved a test MAE of 2.69 for iridium-catalyzed N-allylation and 6.79 for the enantioselectivity of copper-catalyzed allylic alkylation.
- Unified the pipeline on 3 reaction types and demonstrated a classification model distinguishing ligands with a 90% success rate of predicting perfect regioselectivity, broadening its application in drug-product synthesis.
- Empowered multiple senior-level scientists with organic chemistry backgrounds to discover ligands for their own projects.

Research Assistant | Yale University, Victor S. Batista Lab | New Haven, CT

Dec 2020 – Oct 2025

Allostery-aware High-Throughput Drug Discovery and Mechanistic Study of Heterogeneous Catalysis with DFT

- Discovered a novel inhibitor chemical scaffold that uniquely holds the trimer structure of the COVID-19 spike protein by combining high-throughput screening and molecular dynamics simulations; this scaffold can be used to target any coronavirus.
- Detected 3 novel-scaffold hits for GSK3 β from billion-scale, ML-driven virtual drug discovery by establishing an in-house, end-to-end screening platform that converts SMILES strings to hit-enriched candidates via machine-learning-aided molecular docking.

- Identified a new inhibitive allosteric pathway unique to a type of kinase, TYK2 JH2, enabling development of antiallergic drugs with reduced side effects.
- Conceived and co-implemented a LoRA (low-rank adaptation) parameter-efficient fine-tuning strategy to specialize an ensemble of 2,489 domain-expert LLMs for chemical-synthesis prediction, enabling accurate organic-reaction prediction and underpinning a Nature publication (2026).
- Proposed and validated a novel Cu-Pd alloy with nearly 100% Faradaic efficiency for eliminating nitrite waste, proposing a synergistic mechanism that leverages both metals with DFT calculations.
- Revealed a pair of key hydrogen bonds across a human FIMP-TMEM protein pair that were not captured by AlphaFold3, highlighting the value of physics-based refinement.
- Unveiled a novel isomerization mechanism of amine-donated, transition-metal-catalyzed hydrogenation reactions.
- Merged experimental NMR data with advanced models to explore in situ protein conformational changes.
- Mentored students in the School of Engineering to use tight-binding methods to investigate stereoselective polymer formation.
- Collaborated with PIs and scholars across Yale Medical School, Yale School of Public Health, Yale School of Engineering, Yale Center for Molecular Discovery, Yale Center for Research Computing, the University of Michigan, and the Leibniz Institute for Catalysis (Germany).

Research Assistant | University of California, Irvine, Ray Luo Lab | Irvine, CA

Jan 2018 – Jun 2020

Optimized MM-PBSA Method for Accurate and Efficient Drug Candidate Discovery for Membrane Proteins

- Achieved accurate discovery of drug candidates targeting membrane-embedded proteins (e.g. GPCRs) by developing a heterogeneous MM-PBSA free energy calculation method that improved the correlation between calculated and experimental binding free energies from a Pearson r of 0.59 to 0.91.
- Amplified the efficiency of the MM-PBSA method by 65% through GPU implementation and algorithm optimization using CUDA and Fortran, significantly reducing computational time while maintaining high accuracy.

Research Assistant | University of California, Irvine, Sheryl Tsai Lab | Irvine, CA

Jan 2019 – Jun 2020

Molecular docking analyses clarifying ketoreductase regio- and stereoselectivity, enabling rational enzyme engineering

- Achieved 90% concordance between docking-based predictions and experimental outcomes, pinpointing residue mutations (e.g. P94L) critical for stereo-control and identifying two distinct binding motifs in Actinorhodin Polyketide Ketoreductase (ActKR).
- Accelerated project timelines by over 50% by integrating computational feedback loops early in the enzyme engineering workflow.

EDUCATION

- **Yale University**, New Haven, CT — Ph.D. in Computational Chemistry (Advisor: Victor S. Batista) Oct 2025
- **Yale University**, New Haven, CT — M.S. in Chemistry May 2022
- **University of California, Irvine**, Irvine, CA — B.S. in Chemistry, Concentration in Computational & Theoretical Chemistry Jun 2020

PUBLICATIONS

1. Li, H., Sarkar, S., Lu, W., Loftus, P.O., **Qiu, T.**, Shee, Y., Cuomo, A.E., Webster, J.P., et al. Collective intelligence for AI-assisted chemical synthesis. *Nature* (2026).
2. Dan, X., Long, Z., **Qiu, T.**, Menzel, J.P., Shi, Q., Batista, V.S. Nonadiabatic H-atom scattering channels on Ge(111) elucidated by the hierarchical equations of motion. *The Journal of Chemical Physics* (2026).
3. Konieczny, K., **Qiu, T.**, Menzel, J.P., Maslack, J., Batista, V.S., Baráth, E. Hydrogen transfer to internal alkynes using secondary amines on carbon-supported noble metals. *ACS Physical Chemistry Au* (2026).
4. Kyro, G.W., **Qiu, T.**, Batista, V.S. A model-centric review of deep learning for protein design. *arXiv* (2025). arXiv:2502.19173.
5. Long, Z., Meng, J., Weddle, L.R., Videla, P.E., Menzel, J.P., Cabral, D.G.A., Liu, J., ... **Qiu, T.**, et al. The impact of electric fields on processes at electrode interfaces. *Chemical Reviews* (2025).
6. Li, K., **Qiu, T.**, Zhang, B., Ye, Z., Menzel, J.P., Dong, W.J., Pan, Y., Tang, S., Long, Z., et al. Rational pathways tuning facilitates photoelectrochemical upcycling of nitrite to ammonia using a CuPd nanoalloy on a GaN/Si photocathode. *Environmental Science & Technology* (2025).

7. Gisriel, C.J., Malavath, T., **Qiu, T.**, Menzel, J.P., Batista, V.S., Brudvig, G.W., & Utschig, L.M. Structure of a biohybrid photosystem I–platinum nanoparticle solar fuel catalyst. *Nature Communications* (2024). doi:10.1038/s41467-024-53476-y
8. Wang, J., Maschietto, F., **Qiu, T.**, Arantes, P.R., Skeens, E., Palermo, G., Lisi, G.P., Batista, V.S. Substrate-independent activation pathways of the CRISPR-Cas9 HNH nuclease. *Biophysical Journal* (2023). doi:10.1016/j.bpj.2023.11.005.
9. Maschietto, F., **Qiu, T.**, Wang, J., Shi, Y., Allen, B., Lisi, G.P., Lolis, E., Batista, V.S. Valproate-coenzyme A conjugate blocks opening of receptor binding domains in the spike trimer of SARS-CoV-2 through an allosteric mechanism. *Computational and Structural Biotechnology Journal* (2023). doi:10.1016/j.csbj.2023.01.014
10. Zhao, S., Ni, F., **Qiu, T.**, Wolff, J.T., Tsai, S.-C., & Luo, R. Molecular basis for polyketide ketoreductase–substrate interactions. *International Journal of Molecular Sciences* (2020). doi:10.3390/ijms21207562
11. Greene, D.A., Qi, R., Nguyen, R., **Qiu, T.**, & Luo, R. Heterogeneous dielectric implicit membrane model for the calculation of MMPBSA binding free energies. *Journal of Chemical Information and Modeling* (2019). doi:10.1021/acs.jcim.9b00363
12. Wei, H., Luo, A., **Qiu, T.**, Luo, R., & Qi, R. Improved Poisson–Boltzmann methods for high-performance computing. *Journal of Chemical Theory and Computation* (2019). doi:10.1021/acs.jctc.9b00602
13. **Qiu, T.**, & Luo, R. Computational analysis of the dimer interface of p53 bound to DNA. Poster, *ACS National Meeting & Expo, COMP Division* (2019).

RESEARCH FUNDING & AWARDS

- **Yale Ventures Blavatnik Award (FY25)** — preliminary GSK3 β virtual-screening results adopted by a PI into the winning proposal. Feb 2025
- **Yale–Boehringer Ingelheim Thesis Grant** — built the machine-learning catalyst-design collaboration that delivered the results behind the award. Jun 2024
- **MURI Grant (Multidisciplinary University Research Initiative)** — generated preliminary Cu–Pd catalyst-mechanism results a PI used to win and renew the multi-year award. Sep 2024
- Allergan Undergraduate Research Fellowship (awarded to fewer than ten students per year) 2019
- Summer Undergraduate Research Fellowship (SURP), two consecutive years 2018, 2019
- Undergraduate Research Opportunities Fellowship (UROP), three consecutive quarters 2018–19
- Phi Beta Kappa Member (top 2% of juniors by academic performance) 2019
- Dean’s Honor List (8 consecutive quarters) 2016–2020

PROFESSIONAL DEVELOPMENT & SERVICE

- Scientific Reviewer, *Journal of Chemical Theory and Computation* Oct 2023 – Present
- Scientific Reviewer, *Journal of Medicinal Chemistry* Jan 2024 – Present
- Scientific Reviewer, *The American Journal of Undergraduate Research*
- Organizer & Presenter, AI4Science Workshop, Yale Chemistry Department Feb 2025 – Present
- Member, American Chemical Society (ACS), New Haven Local Section May 2020 – Present
- Member, Yale University Chapter of the Biophysical Society Feb 2025 – Present

LEADERSHIP & ADDITIONAL

- Tender Night Student Music Club, UC Irvine — Mixer & Music Producer 2017–2020
- **Languages:** Chinese (Native), English (Fluent / Business), Japanese (Conversational).